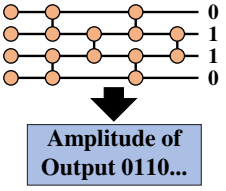
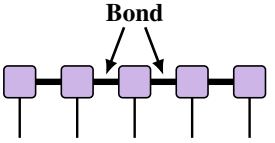
Four Ways to Simulate a Quantum Circuit on a Classical Computer

	State Vector	Amplitude Sampling	Tensor Network	Pauli Propagation
Keeps in Memory	$\alpha_{000\ldots0}$ $\alpha_{000\ldots1}$ $\vdots$ $\alpha_{111\ldots1}$ 2^n Amplitudes All 2^n Amplitudes of the State	 The Circuit as Tensors, Contracted per Output	 The State as Small Connected Tensors	*This Work* Coefficient / Pauli String: 0.35 X Z I; 0.12 Y I Z; 0.08 Z X X A List of Pauli Terms of the Observable
Returns	The Whole State	Amplitudes of the Chosen Outputs	An Approximate State	The Expectation Value of an Observable
Cost Grows With	Qubits: Memory Doubles per Qubit	Entanglement in the Circuit	Entanglement (Bond Size)	**The Number of Terms Kept**
Stops At	About 50 Qubits (40 Qubits = 17 TB)	Deep, Highly Entangled Circuits	Deep Circuits With 2D Connections	Memory for the Terms: Many GPUs Needed

The First Three Follow the Quantum State



Pauli Propagation Follows the Observable: 100+ Qubits, Cost Set by the Terms Kept